

AI text detector · full pilot comparison

AI recall = generated texts detected · Human FPR = human texts falsely flagged · Dashed marker = 2% FPR target

MODEL / TRAIN TIER	AI RECALL · 0-100%	HUMAN FALSE POSITIVES
MIXED TEST631 human + 631 AI · 35% papers in each class		
Qwen3 LoRA · 86.4% cutoff	100.0% (631/631)	0.6% (4/631)
Qwen3 LoRA · 3.7% cutoff	100.0% (631/631)	3.0% (19/631)
Character TF-IDF · full	97.0% (612/631)	2.4% (15/631)
Character TF-IDF · medium	95.2% (601/631)	2.1% (13/631)
Character TF-IDF · small	89.7% (566/631)	2.7% (17/631)
Character TF-IDF · tiny	67.2% (424/631)	1.3% (8/631)
Word TF-IDF · medium	90.8% (573/631)	1.9% (12/631)
MiniLM embedding · medium	30.3% (191/631)	1.4% (9/631)
Load-bearing PR style · frozen	7.6% (48/631)	1.7% (11/631)
EditLens RoBERTa	97.8% (617/631)	1.3% (8/631)
EditLens Llama	99.7% (629/631)	2.5% (16/631)
PAPER ABSTRACT TEST221 human + 221 AI · held-out venues		
Qwen3 LoRA · 86.4% cutoff	100.0% (221/221)	0.9% (2/221)
Qwen3 LoRA · 3.7% cutoff	100.0% (221/221)	4.5% (10/221)
Character TF-IDF · full	99.1% (219/221)	3.2% (7/221)
Character TF-IDF · medium	99.1% (219/221)	2.7% (6/221)
Character TF-IDF · small	98.6% (218/221)	2.7% (6/221)
Character TF-IDF · tiny	95.9% (212/221)	2.7% (6/221)
Word TF-IDF · full	99.1% (219/221)	2.3% (5/221)
MiniLM embedding · full	58.4% (129/221)	2.7% (6/221)
Load-bearing PR style · frozen	0.0% (0/221)	0.9% (2/221)
EditLens RoBERTa	91.9% (203/221)	0.9% (2/221)
EditLens Llama	98.6% (218/221)	1.4% (3/221)
GENERAL EDITLENS TEST2,000 human + 2,000 AI; Llama scored 500 + 500		
Qwen3 LoRA · 86.4% cutoff	99.95% (1999/2000)	0.4% (7/2000)
Qwen3 LoRA · 3.7% cutoff	100.0% (2000/2000)	2.1% (41/2000)
Character TF-IDF · medium	98.9% (1978/2000)	1.9% (39/2000)
Word TF-IDF · medium	96.4% (1928/2000)	2.0% (40/2000)
MiniLM embedding · medium	47.1% (942/2000)	1.6% (31/2000)
Load-bearing PR style · frozen	6.9% (137/2000)	1.6% (31/2000)
EditLens RoBERTa	99.95% (1999/2000)	3.1% (62/2000)
EditLens Llama · small test	100.0% (500/500)	3.6% (18/500)
ENRON EMAIL TEST1,800 human + 1,800 AI; Llama scored 500 + 500		
Qwen3 LoRA · 86.4% cutoff	99.94% (1799/1800)	0.1% (2/1800)
Qwen3 LoRA · 3.7% cutoff	100.0% (1800/1800)	2.1% (38/1800)
Character TF-IDF · general medium	75.9% (1366/1800)	0.4% (7/1800)
Word TF-IDF · general medium	71.0% (1278/1800)	2.1% (38/1800)
MiniLM embedding · general medium	31.7% (571/1800)	0.9% (17/1800)
Load-bearing PR style · frozen	0.0% (0/1800)	0.1% (2/1800)
EditLens RoBERTa	98.1% (1766/1800)	2.7% (49/1800)
EditLens Llama · small test	100.0% (500/500)	1.0% (5/500)
MIXED-TEST PMC SUBSET87 human + 87 AI · papers held out from the Qwen3 mixed training split		
Qwen3 LoRA · 86.4% cutoff	100.0% (87/87)	1.1% (1/87)
Qwen3 LoRA · 3.7% cutoff	100.0% (87/87)	5.7% (5/87)
Character TF-IDF · mixed full	98.9% (86/87)	10.3% (9/87)
Character TF-IDF · mixed medium	98.9% (86/87)	9.2% (8/87)
Character TF-IDF · mixed small	95.4% (83/87)	13.8% (12/87)
Character TF-IDF · mixed tiny	86.2% (75/87)	8.0% (7/87)
Word TF-IDF · mixed medium	98.9% (86/87)	6.9% (6/87)
MiniLM embedding · mixed medium	28.7% (25/87)	1.1% (1/87)
Load-bearing PR style · frozen	0.0% (0/87)	0.0% (0/87)
EditLens RoBERTa	89.7% (78/87)	1.1% (1/87)
EditLens Llama	97.7% (85/87)	4.6% (4/87)
PMC-ONLY ABSTRACT TEST87 human + 87 AI · separate split overlaps mixed training, so Qwen3 is omitted		
Character TF-IDF · full	96.6% (84/87)	0.0% (0/87)
Word TF-IDF · full	97.7% (85/87)	2.3% (2/87)
MiniLM embedding · full	20.7% (18/87)	1.1% (1/87)
Load-bearing PR style · frozen	0.0% (0/87)	1.1% (1/87)
EditLens RoBERTa	92.0% (80/87)	1.1% (1/87)
EditLens Llama	97.7% (85/87)	1.1% (1/87)
SWAPPED-GENERATOR PAPER TEST182 human + 182 AI · same held-out human works, alternate generator		
Qwen3 LoRA · 86.4% cutoff	100.0% (182/182)	0.0% (0/182)
Qwen3 LoRA · 3.7% cutoff	100.0% (182/182)	3.3% (6/182)
Character TF-IDF · paper full	97.3% (177/182)	2.7% (5/182)
Word TF-IDF · paper full	100.0% (182/182)	2.7% (5/182)
MiniLM embedding · paper full	61.0% (111/182)	2.2% (4/182)
Load-bearing PR style · frozen	0.0% (0/182)	0.5% (1/182)
EditLens RoBERTa	90.1% (164/182)	1.1% (2/182)
EditLens Llama	100.0% (182/182)	1.1% (2/182)
HUMAN ACL ABSTRACT AUDIThuman only · paper/mixed audits exclude sampled paper works; general audits have a larger pool		
Qwen3 LoRA · 86.4% cutoff	human only	0.6% (97/17560)
Qwen3 LoRA · 3.7% cutoff	human only	2.2% (383/17560)
Character TF-IDF · mixed medium	human only	1.8% (317/17560)
Word TF-IDF · mixed medium	human only	0.8% (139/17560)
Character TF-IDF · paper full	human only	1.0% (177/17560)
Character TF-IDF · general only	human only	64.8% (11854/18287)
Word TF-IDF · general only	human only	49.1% (8983/18287)
MiniLM embedding · general only	human only	1.4% (259/18287)
HUMAN PMC BODY AUDIT261 human-only text chunks from held-out full papers · false positives only		
Qwen3 LoRA · 86.4% cutoff	human only	1.5% (4/261)
Qwen3 LoRA · 3.7% cutoff	human only	5.7% (15/261)
Character TF-IDF · PMC full	human only	2.7% (7/261)
EditLens RoBERTa · PMC threshold	human only	0.8% (2/261)
Character TF-IDF · general only	human only	34.9% (91/261)

FPR bars use 0–10%; mixed PMC uses 0–15%, ACL audit 0–70%, and PMC body audit 0–40%. Rates are observed, not confidence bounds. Thresholds were selected on each model's validation set. Purple Qwen3 rows use the mixed-validation midpoint; grey rows preserve the original cutoff. PMC-only overlaps Qwen3 training. Paper AI uses two small local generators. Load-bearing scores a GitHub PR writing style, not authorship. EditLens is noncommercial.